STAT452
Ridge and LASSO Regression

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Outline

Limitations of Ordinary Least Squares (OLS)

Ordinary Least Squares (OLS) is widely used in linear regression for estimating model parameters, but it suffers from several important limitations, especially in high-dimensional or noisy data settings:

- **Multicollinearity:**
  - When predictor variables are highly correlated, the design matrix becomes nearly singular.
  - This leads to unstable and highly sensitive coefficient estimates.
  - Small changes in data can cause large fluctuations in parameter values.
- **High Variance:**
  - OLS minimizes residual sum of squares, which may lead to overfitting in presence of many predictors.
  - As the number of predictors (p) approaches or exceeds the number of observations (n), the model's predictions become highly variable.
  - Poor generalization to unseen data.
- **Lack of Variable Selection:**
  - OLS includes all predictors regardless of their relevance.
  - Irrelevant or redundant predictors increase model complexity and reduce interpretability.
- **Low Interpretability:**
  - With many predictors, it becomes difficult to explain how each variable influences the response.
  - Difficult to identify the most significant predictors.

Ridge Regression

Ridge Regression (also known as L_2 regularization) is a technique designed to address overfitting and multicollinearity in linear regression by adding a penalty term to the loss function.

- **Regularization Term:**
  - Adds $\lambda \sum_{j=1}^{p} \beta_j^2$ to the residual sum of squares.
  - λ is a tuning parameter that controls the strength of the penalty.
- **Shrinkage Effect:**
  - Coefficient estimates are shrunk toward zero but never exactly zero.
  - Reduces the impact of less informative predictors.
- **Reduced Variance:**
  - By shrinking coefficients, Ridge lowers the model variance.
  - Leads to better generalization and prediction accuracy on unseen data.
- **Limitation – No Variable Selection:**
  - All predictors remain in the model regardless of their significance.
  - Less interpretable when dealing with high-dimensional data.

LASSO Regression

LASSO (Least Absolute Shrinkage and Selection Operator) is a regularization technique using L_1 penalty, promoting sparsity in the model.

- **Regularization Term:**
  - Adds $\lambda \sum_{j=1}^{p} |\beta_j|$ to the loss function.
  - Encourages some coefficients to become exactly zero.
- **Variable Selection:**
  - Performs automatic feature selection by excluding irrelevant predictors.
  - Enhances model interpretability and simplifies the final model.
- **Comparison with Ridge:**
  - Ridge shrinks coefficients continuously; LASSO can shrink some to zero.
  - Ridge is better when all predictors are relevant; LASSO excels when only a subset matters.
- **Limitation:**
  - May behave unstably when predictors are highly correlated.
  - May select only one variable from a group of correlated ones.

OLS vs. Ridge vs. Lasso

Ordinary Least Squares minimizes:

$$\sum_{i=1}^{n}\left(y_i - \hat{\beta}_0 - \hat{\beta}_1 x_{i1} - \cdots - \hat{\beta}_p x_{ip}\right)^2$$

Ridge Regression minimizes:

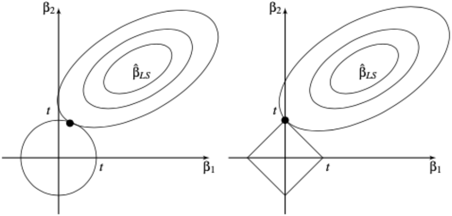
$$\sum_{i=1}^{n}\left(y_i - \hat{\beta}_0 - \hat{\beta}_1 x_{i1} - \cdots - \hat{\beta}_p x_{ip}\right)^2 \quad \text{subject to} \quad \sum_{j=1}^{p}\hat{\beta}_j^2 \leq t$$

Lasso Regression minimizes:

$$\sum_{i=1}^{n}\left(y_i - \hat{\beta}_0 - \hat{\beta}_1 x_{i1} - \cdots - \hat{\beta}_p x_{ip}\right)^2 \quad \text{subject to} \quad \sum_{j=1}^{p}|\hat{\beta}_j| \leq t$$

Note: No constraint is placed on the magnitude of the intercept $\hat{\beta}_0$.

Geometric Illustration of Ridge and Lasso Estimates



- Ellipses are contours of the loss function:

$$\sum_{i=1}^{n}(y_i - \hat{\beta}_0 - \hat{\beta}_1 x_{i1} - \hat{\beta}_2 x_{i2})^2$$

- Centered at the OLS solution: $(\hat{\beta}_{1,\text{OLS}}, \hat{\beta}_{2,\text{OLS}})$.
- **Left:** Ellipse intersects the circle constraint $(L_2) \to$ Ridge estimate.
- **Right:** Ellipse intersects the diamond constraint $(L_1) \to$ LASSO estimate.

Equivalent Forms of Ridge and Lasso

By the **Lagrange multiplier method**, minimizing the residual sum of squares under constraints:

$$\sum_{i=1}^{n}(y_i - \hat{\beta}_0 - \hat{\beta}_1 x_{i1} - \cdots - \hat{\beta}_p x_{ip})^2 \quad \text{subject to} \quad \sum_{j=1}^{p} \hat{\beta}_j^2 \leq t \quad \text{or} \quad \sum_{j=1}^{p} |\hat{\beta}_j| \leq t,$$

is equivalent to the penalized (regularized) forms:

Ridge Regression, minimizing:

$$\sum_{i=1}^{n}(y_i - \hat{\beta}_0 - \hat{\beta}_1 x_{i1} - \cdots - \hat{\beta}_p x_{ip})^2 + \lambda \sum_{j=1}^{p} \hat{\beta}_j^2.$$

Lasso, minimizing:

$$\sum_{i=1}^{n}(y_i - \hat{\beta}_0 - \hat{\beta}_1 x_{i1} - \cdots - \hat{\beta}_p x_{ip})^2 + \lambda \sum_{j=1}^{p} |\hat{\beta}_j|.$$

Ridge Regression: Motivation

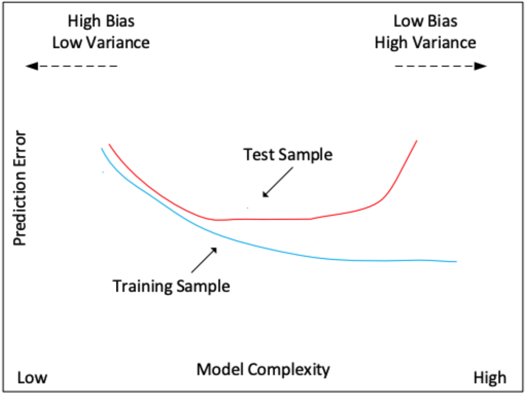
- Ridge regression addresses limitations of OLS, especially when predictors are highly correlated or $p > n$.
- OLS estimates can have large variance and unstable coefficients.
- Ridge introduces a penalty term that shrinks coefficients towards zero, reducing variance and improving prediction.
- It provides biased but lower-variance estimators.

Bias-Variance Trade-Off

- The Mean Squared Error (MSE) is decomposed as:

$$\text{MSE} = \text{Bias}^2 + \text{Variance}.$$

- Increasing model complexity reduces bias but increases variance.
- Ridge finds a balance: it slightly increases bias but significantly reduces variance.



Standardization and Intercept in Ridge Regression

Before estimating the Ridge coefficients, two key assumptions are made:

- **Intercept is not penalized**: The penalty applies only to the slope coefficients $(\beta_1, \beta_2, \dots)$.
- **Predictors are standardized**:
  - Ensures comparability between coefficients.
  - Prevents the penalization from disproportionately affecting variables with larger scales.

Standardization transforms the data as follows:

$$\beta_0 = \bar{y} = \frac{1}{n}\sum_{j=1}^{n} y_j, \qquad \bar{x}_{ij} = \frac{x_{ij}}{\sqrt{\frac{1}{n}\sum_{j=1}^{n}(x_{ij} - \bar{x}_j)^2}}.$$

where $\bar{y}$, $\bar{x}_{ij}$, and $\bar{x}_j$ are the standardized values.

Mathematical Formulation

Ridge regression solves the following penalized optimization problem:

$$\hat{\beta}_{\text{ridge}} = \arg\min_{\beta} \left\{ \sum_{i=1}^{n} \left(y_i - (\beta_0 + \sum_{j=1}^{p} \beta_j x_{ij}) \right)^2 + \lambda \sum_{j=1}^{p} \beta_j^2 \right\}.$$

- The second term is the ℓ_2 penalty.
- λ controls the strength of shrinkage.
- Intercept β_0 is typically not penalized.

Mathematical Formulation

- In matrix notation:

$$\hat{\beta}_{\text{ridge}} = \arg\min_{\beta} \left(\frac{1}{2} \|y - X\beta\|^2 + \lambda \|\beta\|^2 \right).$$

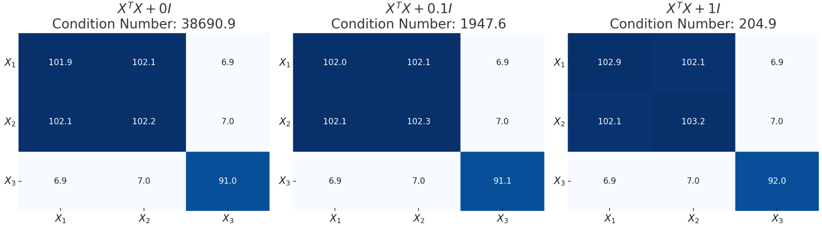
- The closed-form solution is:

$$\hat{\beta}_{\text{ridge}} = (X^T X + \lambda I)^{-1} X^T y.$$

- Adding λI ensures that the matrix is invertible even under multicollinearity.

Effect of Regularization on $X^T X$

- When $X^T X$ is nearly singular due to multicollinearity, inversion is unstable.
- Adding λI stabilizes the inversion process.



Significance of λ

- λ is the tuning parameter controlling shrinkage.
- $\lambda = 0$ gives OLS; $\lambda \to \infty$ shrinks coefficients to zero.
- Chosen using cross-validation to minimize prediction error.
- Affects bias-variance tradeoff and effective degrees of freedom.

Effective Degrees of Freedom

- In Ridge, degrees of freedom is no longer p.
- Defined as:

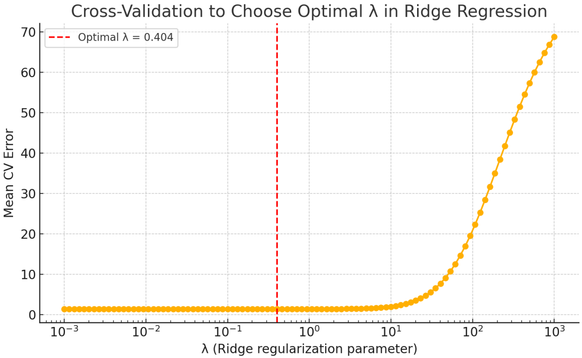
$$df(\lambda) = \mathrm{tr}[(X^T X + \lambda I)^{-1} X^T X] = \sum_{j=1}^{p} \frac{d_j^2}{d_j^2 + \lambda}.$$

- d_j are singular values of X.
- As $\lambda \to \infty$, $df(\lambda) \to 0$.

Cross-Validation for Optimal λ

To select the optimal shrinkage parameter λ, we use cross-validation:

- Try a range of λ values on training folds.
- Compute prediction error on validation folds.
- Choose λ that minimizes average validation error.



Optimal λ found: $\lambda \approx 0.404$

Ridge Estimates Are Biased but Have Smaller Variance

- Recall OLS estimate for $\beta = (\beta_0, \beta_1, \ldots, \beta_p)^T$ is:

$$\hat{\beta}_{\text{OLS}} = (X^T X)^{-1} X^T Y.$$

- Ridge estimate is:

$$\hat{\beta}_{\text{ridge}} = (X^T X + \lambda I_p)^{-1} X^T Y.$$

  - Assumes X is standardized: mean 0 and variance 1 for each predictor.

- Expected value of Ridge estimate:

$$\mathbb{E}[\hat{\beta}_{\text{ridge}}] = (I_p + \lambda X^T X)^{-1} \beta \neq \beta.$$

  $\Rightarrow$ Ridge estimate is **biased**.

- If predictors are standardized and uncorrelated:

$$\hat{\beta}_{j,\lambda,\text{Ridge}} = \frac{1}{1+\lambda} \hat{\beta}_{j,\text{OLS}}$$

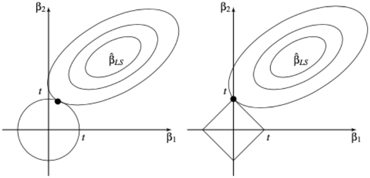
- Variance of Ridge estimate is **smaller** than that of OLS, especially effective when data has **multicollinearity**.

Why Ridge Shrinks But Keeps All Variables

- Ridge Regression uses an L_2 penalty: $\lambda \sum_{j=1}^{p} \beta_j^2$
- The penalty discourages large coefficients by shrinking them towards zero.
- However, L_2 norm does not allow coefficients to become exactly zero.

Geometric Intuition

- The constraint region for Ridge is a **ball** (circle or sphere).
- Ellipses of residuals (RSS contours) typically touch the constraint at non-zero values.
- Hence, all predictors are retained in the model with reduced magnitudes.



Mathematical Formulation

LASSO minimizes the following objective:

$$\hat{\beta}_{\text{lasso}} = \arg\min_{\beta} \left\{ \sum_{i=1}^{n} \left[y_j - \left(\beta_0 + \sum_{j=1}^{p} \beta_j x_{ij} \right) \right]^2 + \lambda \sum_{j=1}^{p} |\beta_j| \right\}$$

Interpretation:

- λ is the shrinkage parameter.
- L_1 penalty: $\|\beta\|_1 = \sum |\beta_j|$.
- Different from Ridge (which uses L_2 penalty).
- LASSO can shrink some coefficients exactly to zero.

Matrix Formulation

LASSO can also be expressed as:

$$\hat{\beta}_{\text{lasso}} = \arg\min_{\beta} \left(\frac{1}{2} \|y - X\beta\|^2 + \lambda \|\beta\|_1 \right)$$

- Provides a unique solution when $X^T X$ is full rank.
- No closed-form solution, because the constraint given by the L_1 penalty is in absolute value, which cannot be differentiated.
- The solutions for the lasso problem are nonlinear in y_i because the constraint has a non-smooth nature.

Properties of Lasso Estimates

- No closed-form formula for the Lasso estimates
- Also biased (toward 0)
- Smaller variance than OLS estimates
- Does **not** perform as well as Ridge when data have **multicollinearity** problem
- Greatest advantage of Lasso: **Sparsity**

Sparsity of Lasso Estimates

- In a model with many predictors:

$$Y = \beta_0 + \beta_1 X_1 + \cdots + \beta_p X_p + \varepsilon.$$

- We may believe many of the β_j's are actually 0.
- Hence, we seek a set of sparse solutions.
- Lasso estimates will set some coefficients exactly equal to 0 when λ is large (or t is small).

So the LASSO will perform model selection for us!

How to Choose λ?

- We need a disciplined way of choosing λ
- Obviously want to choose λ that minimizes the mean squared error
- This issue is part of the bigger problem of **variable selection**

Choosing λ Using Cross-Validation

- A good model should predict well on new data
- Split data into two parts — **training data** and **test data**
- For each λ:
  - Use the training set to train the model
  - Use the model to predict values in the test set
  - Compute the root mean square error (RMSE):

$$\text{RMSE} = \sqrt{\sum_{\text{test data}} (y_i - \hat{y}_i)^2 / n}.$$

- Choose the λ that has the smallest RMSE
- Randomly select training/test sets. Optionally, use multiple splits and average the RMSEs.

Ridge Regression in R

See R-Markdown document . . .